

BASICS, APPLICATIONS AND FUTURE OF ADDITIVE MANUFACTURING TECHNOLOGIES: A REVIEW

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ABSTRACT

This article presents a review work on additive manufacturing technologies as reported in various journals and proceedings. Additive manufacturing technology enables to fabricate physical part directly from the computer aided design data by layer by layer deposition of raw material in solid, liquid and gas form without using any tools, and without producing any waste material. In its early days, this technology was mainly used to build up prototype models in various industries. However, current application of additive manufacturing is not limited around prototyping as it is moving rapidly towards the fabrication of functional end-user parts. It has shown huge potential in the field of tooling, manufacturing and medicine for direct or indirect fabrication purposes. Moreover, additive technologies require a lot of research and development to solve issues like speed, materials, non-linear behavior, and high cost of the additive manufacturing systems.

Keywords: Additive manufacturing, computer aided design, fabrication, product

1. INTRODUCTION

In the early days, rapid prototyping was the term used to describe a group of emerging material addition technologies, started in the mid-1980s with the debut of 3D Systems Inc.'s (Valencia, CA) SLA-1, with the idea of fabricating prototypes of products directly from computer aided design (CAD) data (Cheah et al., 2005; Jee and Sachs, 2000). Since then this technology has shown significant improvement in speed, part accuracy and material properties and also has widened the range of its application areas. Moreover, attracting features and widely acceptance of this emerging technology has lead to use the term additive

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manufacturing (AM) by the manufacturing industry and 3D printing by the general public rather than using the term rapid prototyping (Campbell et al., 2012). Recently ASTM (American Society for Testing and Materials) standard F-2792 defines it as: “Additive manufacturing (AM) is a process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to subtractive manufacturing technologies” (ASTM International, 2012).

The major applications of this technology include the fabrication of various kinds of models and prototypes, which are used for visual inspection, concept evaluation and presentation and functional testing in various stages of the product development process (Noorani, 2006), and also serve as an indispensable tool for shortening product design and development time cycles (Cheah et al., 2005). The impact of rapid prototyping on product development process (Noorani, 2006) can be seen in Figure 1. Pham and Gault (1998) reported that use of rapid prototyping can reduce new product costs by up to 70% and the introduction time to market by 90%. In addition to prototyping, it has shown its potential in other sectors also such as rapid manufacturing (RM) for the manufacturing of end user parts in small batches, rapid tooling (RT) for the fabrication of manufacturing tools or moulds (Giannatsis and Dedoussis, 2009) and medicine (Rengier et al., 2010).

AM technology can be divided into two phases (Da Rosa et al., 2004): virtual and physical. Before the fabrication of physical models, firstly comes the virtual phase in which 3D CAD model is prepared by using CAD software packages such as Pro-E, Catia etc. The fabrication of the physical model is the second phase in which suitable AM process uses CAD data to develop physical models.

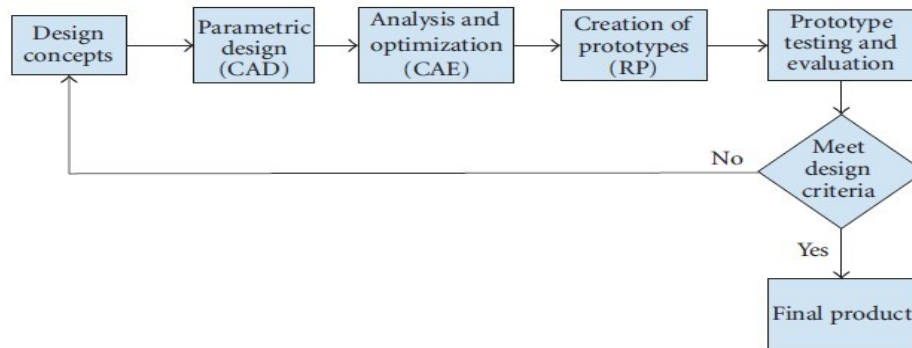


Figure 1. Product development cycle (Noorani, 2006).

Some of the common AM processes used worldwide in the fabrication process are stereolithography (SL), selective laser sintering (SLS), fused deposition modeling (FDM), 3D printing (3DP), laminated object manufacturing (LOM), polyjet printing (PP), electron beam melting (EBM) and laser engineered net shaping (LENS). The present paper deals with only the above mentioned AM processes. All these processes have a common feature (Yan and Gu, 1997): the model is fabricated by adding layer by layer of raw materials, rather than removing or deforming materials as in conventional manufacturing processes. Additionally, the unique characteristic which makes this technology more special (in comparison to conventional manufacturing) is its ability to fabricate physical parts directly, i.e., it does not require any change in tool, fixture and other related activities for fabrication purposes (Choi and Samavedam, 2001).

As mentioned earlier, these processes were first developed to produce models (prototypes), but the gradual improvement of AM technology (speed, accuracy and material) made it possible to expand its applications in other industrial areas (Giannatsis and Dedoussis, 2009). Wohlers associates conducted a survey amongst various manufacturers and lead users to investigate different application areas of AM technologies, and received feedbacks are shown in Figure 2 (Wohlers, 2009). It shows the diverse applications of these emerging technologies. Whereas Figure 3 presents the growth of AM technology for the direct fabrication of finishing products; it has been increased from 3.9% to 24% in last nine years (Wohlers, 2012). The objective of the present study is to review the most common available AM processes, their popular applications, related issues and future of this emerging technology.

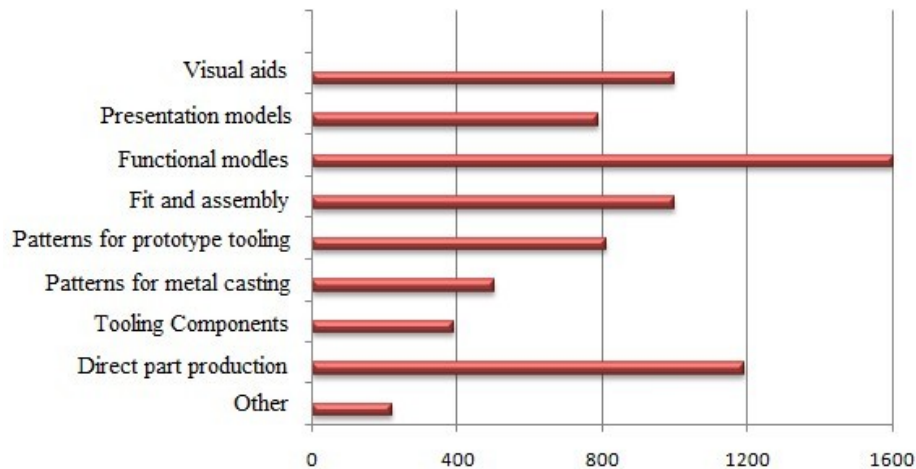


Figure 2. Different applications for different AM technologies (Wohlers, 2009).

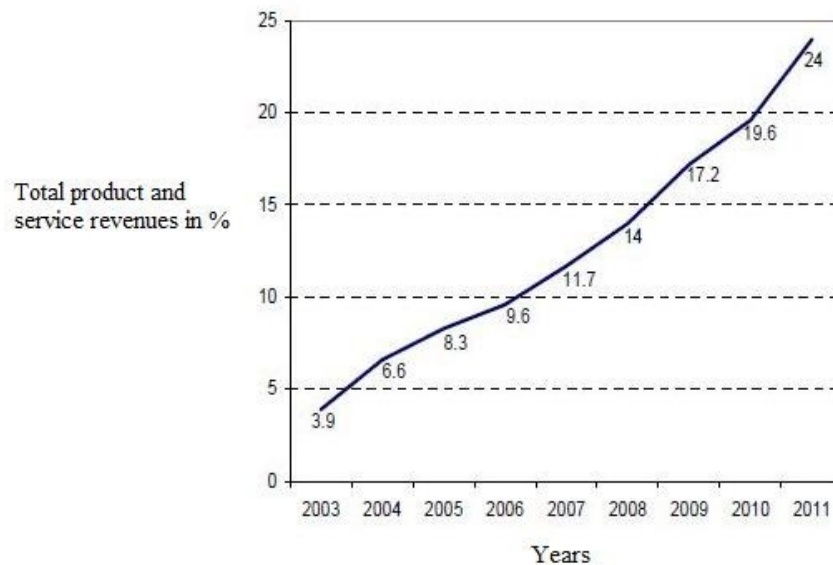


Figure 3. Growth of AM in direct fabrication of finished products (Wohlers, 2012).

2. TECHNOLOGY OVERVIEW

AM technology can be classified with two major aspects, first is the type of material they used (solid, liquid and powder form) and the second is the deposition technique for part fabrication. Each process has its own capabilities, limitations and applications in fabricating prototype models. There are currently a number of AM processes available in the market, based on special sintering, layering or deposition methods. A comparison of selected AM technologies is presented in Table 1. The basic process of additive fabrication and different available AM processes are precisely discussed in below sections.

2.1. The Basic Process of AM

The first step for all AM processes is common, which is 3D geometric modeling, in which a digital CAD model of required geometry is designed by using CAD software package such as Pro/E, Catia. The CAD file is converted into standard triangulation language (STL) file format that is a standard input file format for almost all AM systems. The STL file is imported into specific software (Magics RP by Materialise, and Viscam by Marcam), where it is pre-processed. The part model is oriented for building and a support structure is provided for the part's surfaces that are facing down. After selecting the optimal part orientation, model is sliced at distances equal to the layer thickness. Additional information is provided with regard to the design of supports (when required), the part orientation and the scan path for each layer.

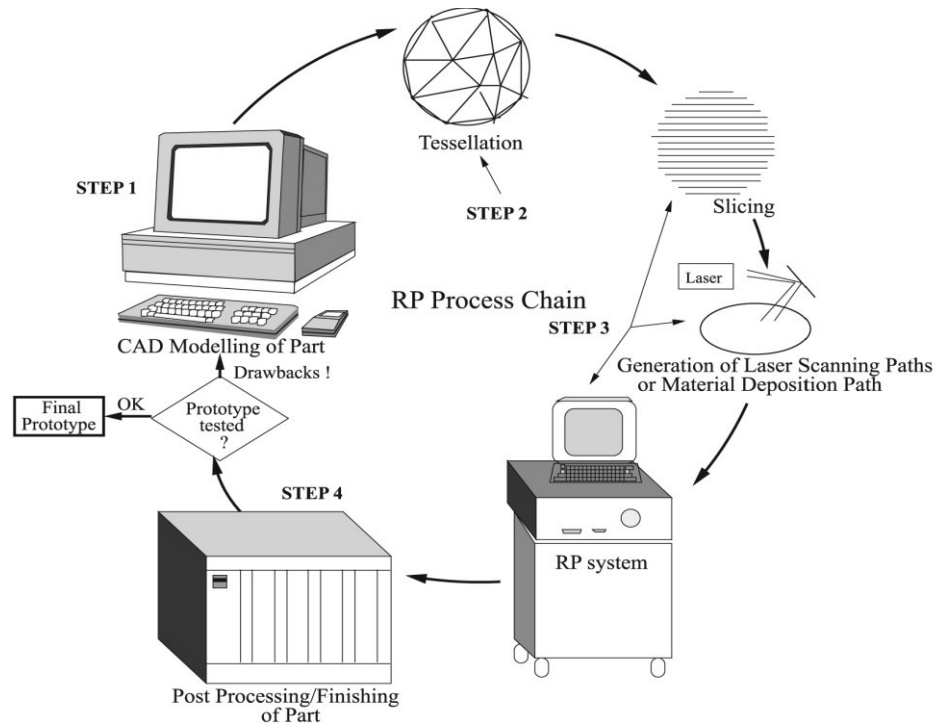


Figure 4. Process chain involved in AM processes (Pandey et al., 2003).

This information about each section is then electronically transmitted to the AM system.

The physical part is fabricated using one of the many currently available AM systems by adding material layer by layer rather than removing it. The final step is post-processing. This includes removing the physical part from the platform, detaching the supports and sometime part requires minor cleaning and surface treatment operations to improve its appearance and strength (Petrovic et al., 2011; Rapid Prototyping Primer [online]). Figure 4 explains the general steps involved in AM processing.

2.1.1. STL File

STL (standard triangulation language) file format was developed by 3D Systems for its stereolithography apparatus (SLA) machines in 1987 (Noorani, 2006), and since then it has become the standard input format for almost all RP systems. STL file involves the tessellation of the surface and triangulation, which generates a large number of triangular facets to approximate the virtual CAD model as shown in Figure 5.

Table 1. Features of commercially available AM technologies (Chua et al., 2003; Levy, 2010; Mousah, 2011)

Technology	Process	Commercially Available since	Energy	Geometry	Metal	Plastic
SLA	Photocuring	1987	Laser	Scan		☐
SLS	Sintering of powders	1991	Laser	Scan	☐	☐
FDM	Melt extrusion	1991	Extrusion	XYZ Kinetic		☐
3DP	Ink-jet printing	1998	Binder	XYZ Kinetic	☐	☐
LOM	Paper lamination	1990	Laser	Scan		☐
LENS	3D laser cladding	1997	Laser	XYZ Kinetic	☐	
EBM	Sintering of powders	2001	Electron laser beam	Scan (magnetic)	☐	
PP	Ink-jet printing	2001	UV light	XYZ Kinetic		☐

Which means the surface of the model is represented by a set of small triangles. Increasing the number of triangles improves the approximation, but it causes larger file size. Each triangle is described by an outward normal and the coordinates of three points. In short, an STL file consist a list of x, y, and z coordinates triplets that represent a connected set of facets and also includes the direction of the normal vector for each triangle. The output of STL file formats can be expressed in ASCII (American Standard Code for Information Interchange) and Binary format. If the structure of STL file is incorrect such as a large amount of undesirable gaps, facets degeneracy and flipped normal, then some post-processing tasks (manual operations) are obviate to repair theses facet defects (Ma et al., 2001; Tie et al., 2005; Pandey et al., 2003a). Other types of alternate translators ((Noorani, 2006; Wong and Hernandez, 2012) are initial graphic exchange specification (IGES), hewlett-packard graphics language (HPGL), stereolithography contour (SLC), SLI from 3D Systems, CLI from EOS,

stereolithography contour from Stratasys, and F&S from Fockele and Schwarze and computerized tomography (CT) data used for medical imaging.



Figure 5. A CAD model (left) converted into STL file format (right) (Gibson et al., 2010).

2.2. Stereolithography

Stereolithography was introduced by 3D Systems in 1987 (Gibson et al., 2010), which is the first commercially available rapid prototyping system worldwide and still the most widely used process. Stereolithography apparatus (SLA) system as presented in Figure 6 consists of a build platform (elevator), which is mounted in a vat filled with liquid monomer resin (acrylic or epoxy resin), recoating blade and an ultraviolet helium–cadmium laser or argon ion laser.

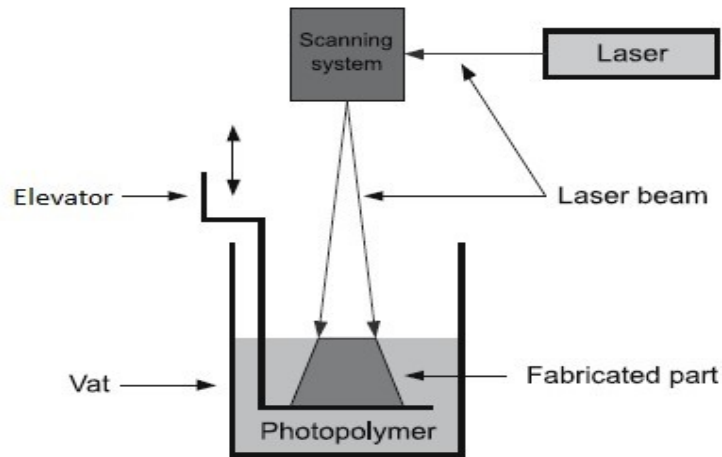


Figure 6. Schematic of SL process (Liu et al., 2005).

In SL process the surface layer of monomer resin is cured selectively by the laser beam, which (laser) follows the same path as generated in sliced model. After creating one layer, the movable platform is lowered into the vat and a fresh liquid resin is spread over the last solidified layer and the whole process is repeated. This process continues layer by layer until the part fabrication is completed. After the whole part fabrication, it is removed from the platform and then removal of the support structure to get the green part. Post-curing of green part is done in a post-curing apparatus (PCA), which is either a controlled furnace or an ultraviolet oven. Post-curing is required to completely solidify the prototype. The advantages

are smooth surface finish, high accuracy whereas disadvantages of this process are expensive and toxic material, time-consuming post-processing and need support structures which may adversely affect the surface finish when removed (Chockalingam et al., 2008; Gibson et al., 2010; Yan and Gu, 1997; Zhou et al., 2000).

2.3. Selective Laser Sintering

Selective laser sintering process was developed at the University of Texas at Austin in 1986, and commercialized by DTM Corporation (Wang et al., 2006), has become popular due to the fact that the SLS process fabricates the parts without using support structure and has the ability to use a variety of powder materials. Materials (Gibson and Shi, 1997; Singh et al., 2012) that can be fabricated in SLS include: polycarbonate (PC), nylon, nylon/glass composite, wax, ceramics, trueform (TM), elastomeric and metal-polymer powders. The SLS process involves fabricating solid parts by fusing powdered materials with the help of a CO₂ laser beam. A very thin layer of powder material is spread with a roller over the part build surface and preheated to a temperature slightly below its melting point. A laser beam follows the cross-section on the powder surface to selectively sinter and bond it in order to produce a layer of the part as presented in Figure 7. After completing one layer, successive layers of powder are deposited and sintered until the whole part is completed. The advantages include the ability to use a variety of thermoplastic powders, easy post-processing, no support structure required. Disadvantages are the abrasive surface of sintered models and high costs of this method (Juster, 1994; Williams and Deckard, 1998; Yang et al., 2002).

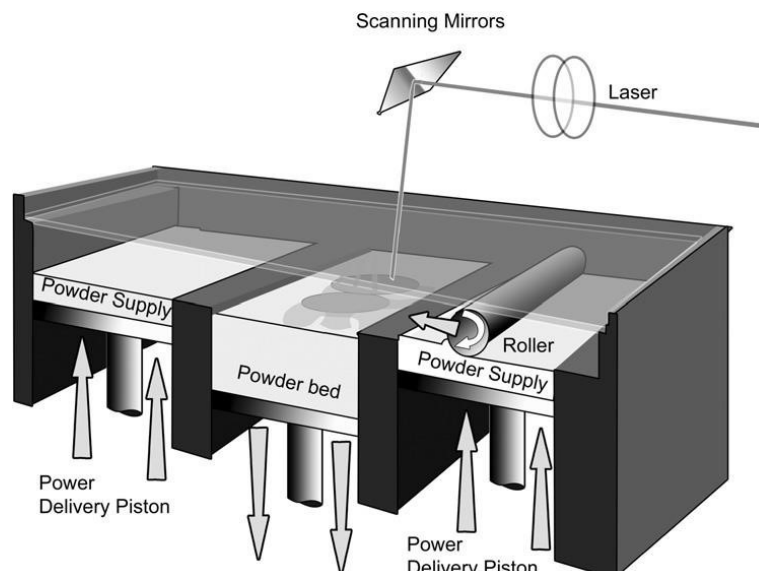


Figure 7. Schematic of SLS process (Goodridge et al., 2012).

2.4. Fused Deposition Modeling

Fused deposition modeling was introduced by Stratasys in 1991 (Noorani, 2006). FDM system is an XY plotter device, which uses a fine nozzle to force out a thermoplastic filament material in a semi-molten form onto a platform in a layer by layer manner. The maintained

lower temperature of the platform assists the extruded material in rapid hardening and solidification. As each layer is extruded, it bonds to the previously built layer and solidify. The latest FDM system includes two nozzles; one for the part material and one for the support material as represented in Figure 8. Once the fabrication is completed, the support structures are removed from the part for the final processing. This process uses a variety of modelling materials and colors, such as acrylic butadiene styrene (standard and medical grade), elastomer, polycarbonate, polyphenolsulfone, and investment casting wax. Advantages of this method are compact construction of the FDM machine, the possibility to sterilize these models and better geometric inaccuracy on the other hand disadvantages are slow building speed and lower surface quality than the stereolithography technology (Lee et al., 2007; Liu et al., 2005; Zein et al., 2002).

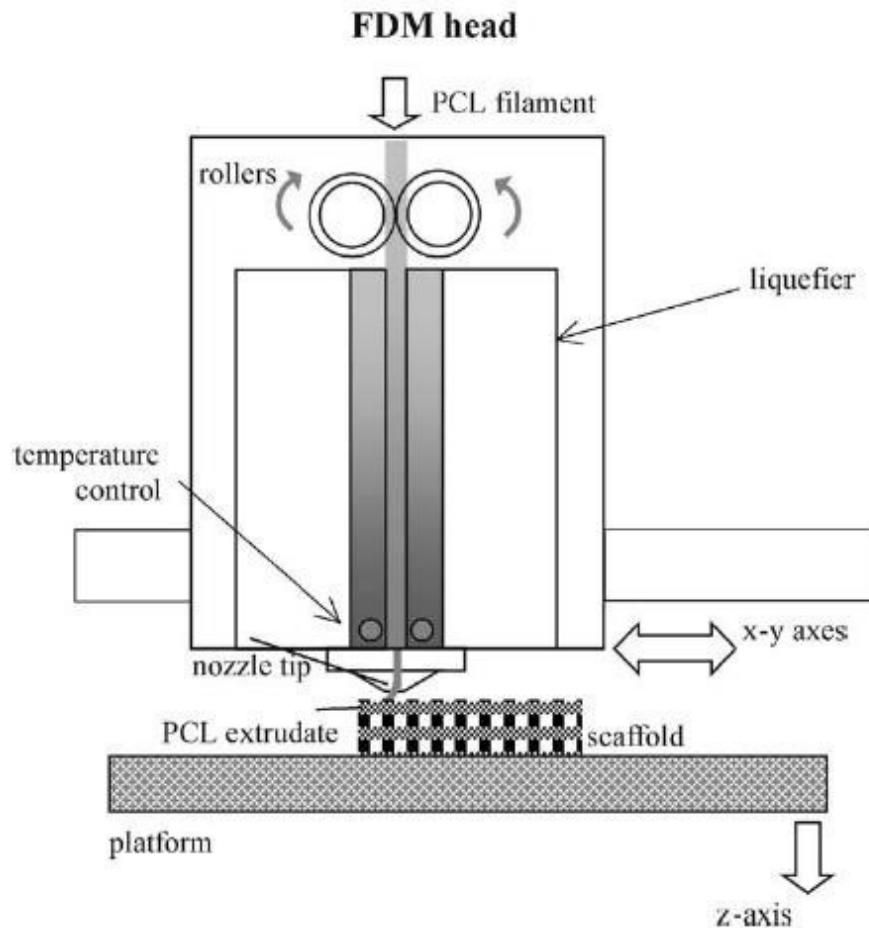


Figure 8. Schematic of FDM process (Zein et al., 2002).

2.5. 3D Printing

3D printing technology developed at Massachusetts Institute of Technology (MIT) and introduced by Solidscapes (Noorani, 2006), Figure 9 shows a schematic of this process. 3DP uses inkjet printing technology to fabricate solid parts. An ink-jet printing head is used to spray the binder material in selective areas to blend the powder together. Once a layer is deposited, the piston that supports the powder bed and the part lowers, so that the next powder

layer can be spread and selectively joined. This layer-by-layer process repeats until the whole part is fabricated. The whole process is completed, unbound powder is removed from the green part and suitable post-processing is carried out.

To increase durability and strength of the part, further it can be infiltrated with epoxy resin, wax, or other materials. This process does not require external supports during the fabrication process since the unsintered powder supports the overhangs. 3DP's advantages are good fabrication speed and low materials cost. Disadvantages are surface finish, moderate strength, hand free post processing and availability of materials (Gill and Kaplas, 2011; Liu et al., 2005; Vaezi and Chua, 2011).

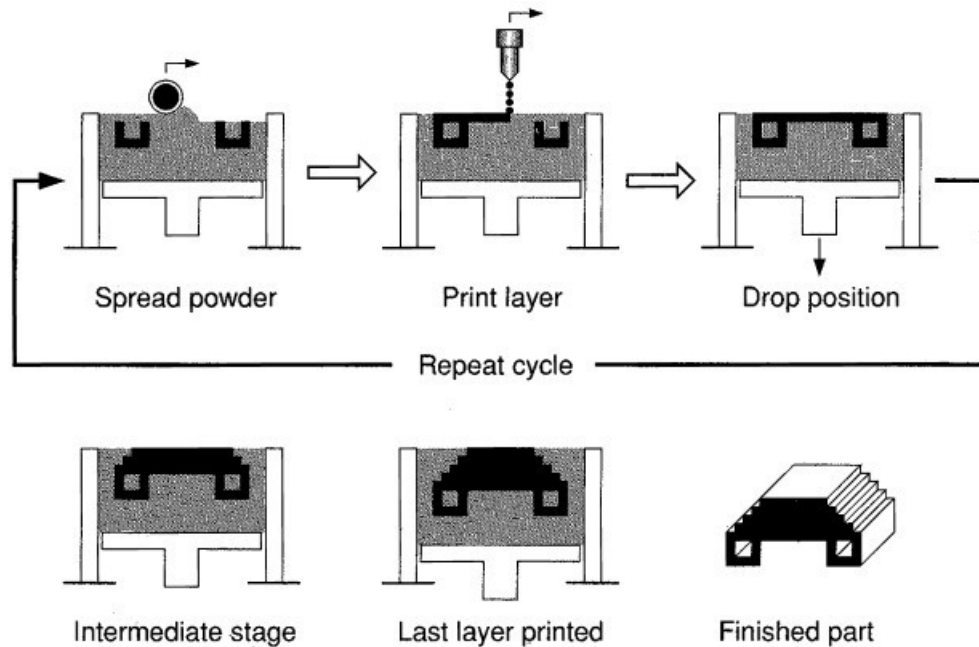


Figure 9. Sequence of 3DP process (Jee et al., 2000).

2.6. Laminated Object Manufacturing

Helisys introduced the laminated object manufacturing; this process is based on the principle of lamination. This technology fabricates parts from paper, ceramic, polymer or metals followed by a subtractive step which cuts the outline with a CO₂ laser cutter to match the cross-section of the model. After each outline has been cut the roll of paper is advanced, then a new layer is glued onto the stack, and the same process is repeated until the whole part is fabricated and then some trimming, hand finishing and curing are required. Those models are fabricated with paper material; they must be painted and sealed to avoid moisture damage. Figure 10 demonstrates the mechanism of a LOM machine. Advantages include no post curing, no support structure needed and well suitable for complex and large parts whereas disadvantages are poor surface finish and environmental issues (not suitable in wet or humid conditions) (Noorani, 2006; Liu et al., 2005).

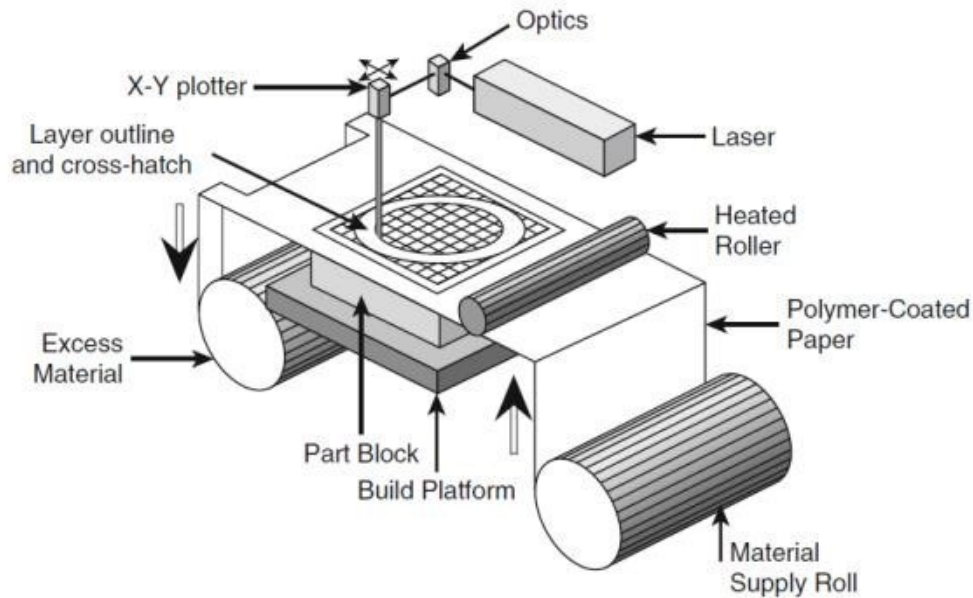


Figure 10. Schematic of LOM process (Yan and Gu, 1997).

2.7. Polyjet Printing

Polyjet printing is based on the MIT's ink jet technology to manufacture physical models; it was patented (Singh, 2011) in 1994. The process consists only of ultraviolet (UV) bulbs and photopolymer materials. In this process, each sprayed photopolymer layer is cured by UV light and produced fully cured solid parts. Two different photopolymer materials are used for building, one for the fabrication of actual model and another gel like material for supporting overhang features. Figure 11 shows schematic of PP process. Supporting material can be easily separated from the part by using a water jet or hand and brush. Finished models do not

require any special baths or extra finishing treatments. But the solid parts fabricated by this method are weaker than others like stereolithography and selective laser sintering (PolyJet Technology [online]; Singh, 2011; Wong and Hernandez, 2012).

2.8. Electron Beam Melting

The EBM process uses the same methodology for fabrication purposes as the SLS process; there is only one EBM based technology available and which is patented by Arcam, a Swedish technology development company. In EBM the energy is provided by the impact of a focused beam of electrons (high voltage 30-60 KV) to melt the powder as presented in Figure 12.

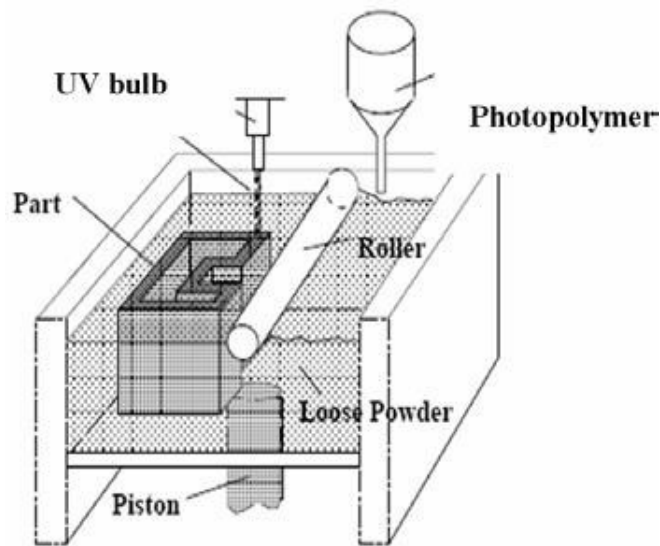


Figure 11. Schematic of PP process (Singh, 2011).

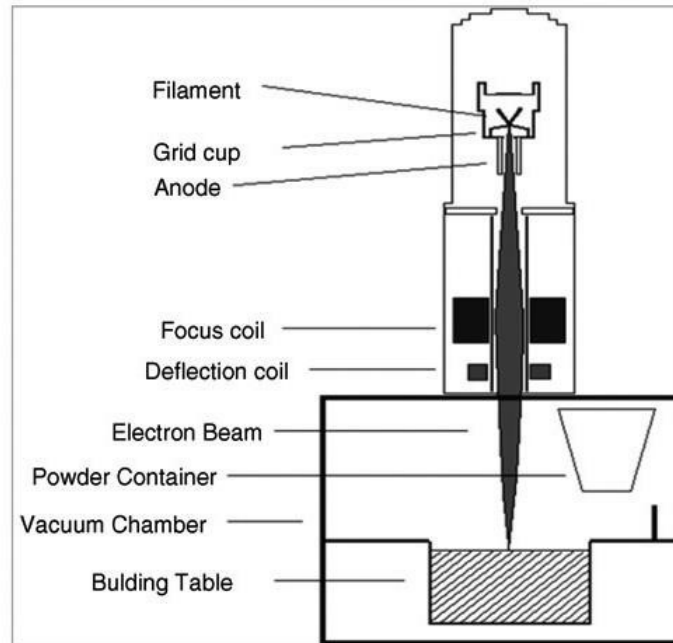


Figure 12. Schematic of EBM process (Singh, 2011; Petrovic et al., 2011).

The electrons are emitted from a filament made of tungsten. The process takes place in a high vacuum chamber to avoid oxidation issues (in case of metal parts) and also vacuum prevents the beam from being deflected and maintains it well on the target. This process can use any material that has an adequate electrical conductivity. However, as the literature suggests till now, only three types of titanium (Ti6Al4V, Ti6Al4V ELI and Ti Grade 2) and cobalt chromium (CoCr ASTM F75) materials are officially released for the EBM process (Petrovic et al., 2011; Wong and Hernandez, 2012). EBM advantages include superior

material properties, excellent accuracy and good build speed where as disadvantages are high power consumption and need to maintain a vacuum chamber (Chua et al., 2003).

2.9. Laser Engineered Net Shaping

Sandia National Laboratories introduced a new technology named as laser engineered net shaping, which can fabricate metal parts directly from CAD file and its capability for rapid manufacture of fully dense metal parts makes it more popular over other AM processes. In this process a high power laser is used to melt metal powder, a deposition head is used to supply the powder material coaxially to the focus of the laser beam as shown in Figure 13. The laser beam is focused to a small spot with the help of one or more lenses. The XY table moves in raster fashion to fabricate each layer of the object and head move up vertically after completing each layer. Metal powders are supplied and distributed either by using pressurized carrier gas or gravity. The complete process occurs in an inert gas chamber to protect the melt pool from atmospheric oxygen for better control of properties. This process can use a variety of materials such as stainless steel, copper, aluminum, etc., as well as composite and functionally graded materials. This technology has the ability to fabricate large, fully-dense metal parts with better metallurgical properties at reasonable speed (Laser engineered net shaping [online]; Worldwide Guide to Rapid Prototyping [online]).

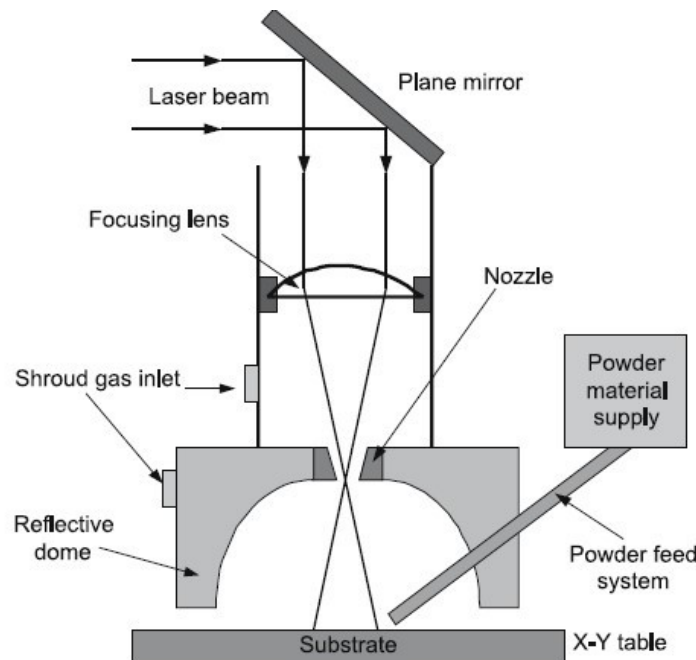


Figure 13. Schematic of LENS process (Liu et al., 2005).

3. ADVANTAGES, CHALLENGES AND FUTURE OF AM TECHNOLOGIES

Some of the most important advantages of AM include (Onuh and Yusuf, 1999; Petrovic et al., 2011):

1. Reduced lead time (per component and product to market), material wastage and costs (direct manufacture savings and mistake avoidance). It was observed that there is an average savings of 80% both in cost and time over the traditional methods. In metal sector, it was observed that the raw material wastage is reduced by up to 40% when using AM processes instead of subtractive (machining) processes.
2. Improved the quality of prototyping. The quality of the parts produced from these methods, all satisfy the requirements for prototype testing and a significant improvement in the mechanical properties and surface characteristics were noticed in comparison to vacuum casting parts.
3. Parts with complex geometries. With the help of this new technology it is now possible to fabricate complex shaped parts of almost any geometry as well as customize parts, as it requires only CAD data for fabrication purposes.
4. No tools, moulds or punches are needed. AM fabricates parts directly from its CAD data without any tooling and without human intervention.

However, this technology also faces a set of issues, and these certain issues have restricted the use in certain application areas. Presently major limitations of AM are build speed, accuracy, part orientation, material properties, surface finish, preprocessing, postprocessing and system cost. Some of the major issues as per available literature (Campbell et al., 2012; Kochan et al., 1999; Onuh and Yusuf, 1999; Wong and Hernandez, 2012) are addressed in below section.

1. Confusions due to its original name “rapid prototyping. At the present time, technologies of rapid prototyping are not just used for prototyping, they have been successfully used in producing finished products but still it is still wrongly considered that this technology is valid only for prototype fabrication.
2. Nonlinear behavior of AM fabricated parts (different resolutions for xyz axes). AM fabricated parts behave (basically strength) different in different directions in comparison to conventionally manufactured parts. They show more strength when the load is applied along the direction of the layer as compared to the build-up direction.
3. The “stair-stepping” phenomenon. As the slicing layer is rectangular in cross section it cannot conform to the curved surface across its entire thickness, as a result it affects surface quality and roughness of produced parts. This phenomenon has been termed as the “stair-stepping” phenomenon.
4. Layer thickness selection. As we know smaller the layer thickness, the greater the resolution of the final part, but small layers results longer data processing time, larger data files and especially a longer build time. This is one important parameter which can be optimized or variable layer thickness may be used over different ranges of the part
5. Use of support structures. Material used for the support structure cannot be recycled and sometime support structures affect part quality, it should be minimized by selecting an optimal build-up orientation.
6. Direct manufacturing of metal parts. This is main challenge existing now. Processing of metal parts directly built with AM technology is still rare in comparison to indirect

methods of AM technology. So AM technology should be extended to build metal parts or prototypes with adequate strength and accuracy

7. Development of new materials. Materials commonly used in present AM systems are polymers, paper and ceramic based. However, there are many interesting materials that need to be focused (such as magnesium, copper, biodegradable polymers, etc.).
8. AM system cost. It is also one of the major issues of concern. Presently AM systems are very costly due to high manufacturing cost or availability of less number of potential users. This kind of situation always occurs at the beginning stage of any new high-technology. But if the manufacturing cost gets reduced or the number of potential users becomes larger, only then the system cost can be reduced. Recently sum low cost AM systems have been introduced in market by just aiming at entry level users are mainly for personal use and thus for high-volume sales of these systems has been noticed.

Now what will be the future this emerging technology, right now it can only be predicted on the basis of ongoing research and development in the field of AM. Few possible predictions reported by Campbell et al., (2012) are:

1. It can be predicted that low or medium cost AM systems will be available in the next ten years as the primary patents are going to expire, and that these AM systems will be easily available to the public, which will result more potential users, demands and the entry of major suppliers to the market with their new and improved materials.
2. One major issue to be focused is speed, and in future if fabricating speed significantly improved with the progress of materials and design process. Physical parts will become available in minutes rather than hours. AM systems are likely to be seen in shopping malls and other locations where consumers can place an order and get their part on the counter in just a very short span of time.
3. Future AM machines with the association of other technology (hybrid AM technology) are not going to be as versatile as the existing AM, due to the reason that these future systems will be specifically developed for a particular product specification, like turbine blades or wing struts.
4. AM systems (not based on all existing technologies) in the future will have the capability of processing multiple materials in comparison to current available AM systems, due to that it will be more easier to fabricate interesting products, presently those are extremely difficult to manufacture using conventional technology.
5. AM systems have already shown its potential in the medical sector, but one area to be the focus is tissue engineering. Where AM associate with new biochemical approaches can be used to generate generating implants with special geometrical characteristics such as scaffolds for the restoration of tissues.

4. APPLICATIONS

4.1. Aerospace and Automotive Industry

AM has emerged as a key enabling technology with its ability to reduce product design and development time cycle effectively, and this feature of this technology has exploited by

automated manufacturer to introduce their new products into the market quickly and in a predictable manner. Aerospace sector (Campbell et al., 2012) has shown interest in these technologies because of the ability to fabricate direct metal parts such as from titanium (suitable for aircrafts), and the ability to fabricate complex and high performance products easily without any tooling. EBM and SLS are now employed in aircraft and aerospace industries. The main goal (Wong and Hernandez, 2012) of automotive and aerospace industry is to manufacture light weight vehicle or aircraft and these AM based technologies are very much able to manufacture lightweight parts as presented in Figure 14. There are lots of examples of the use of AM technology in the aerospace industry. Some of them are design verification of an airline electrical generator, production of the casting pattern of an impeller compressor shroud engine component and fabrication of flight-certified production castings. Where as in automotive application AM has been successfully used for prototyping complex gearbox housing for design verification, fabricating cast metal engine block and creating the production tool of a rear wiper-motor cover (Chua et al., 2003). Petrovic et al., (2011) reported that some empirical studies had performed in the past to evaluate traditional manufacturing processes such as gravity die casting, cross drilling, electrochemical deburring and hole blanking with AM processes for fabricating aerospace and automotive parts. It was observed that when the same part fabricated through AM, had the same mechanical properties and used 40% less material. Another study (Raja et al., 2006) has reported that AM technologies will provide aerospace sector the following benefits:

- Reduction in product introduction lead-time by up to 30% to 70%; □ Savings in non-recurring product introduction costs by up to 45%; and
- 30% to 35% reduction in manufacturing costs for low volume service parts.

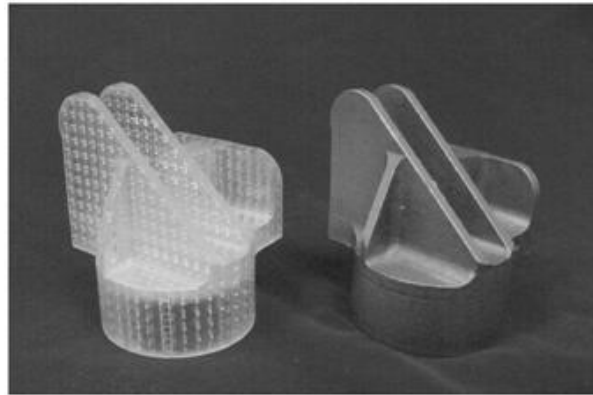


Figure 14. Aerospace component (left) quickcast pattern (right) stainless steel casting (Cheah et al., 2005).

4.2. Artistic Industry

The use of AM technology in jewelry and arts offers new possibilities to design most complex designs or shapes as shown in Figure 15. SLA (from 3D Systems) has been used successfully to develop fine jewelry models (Chua et al., 2003) because of the high printing resolution. Models fabricated by SLA were used as master patterns to create the rubber molds for manufacturing wax patterns. These were later used in the investment casting process to fabricate functional end user product.

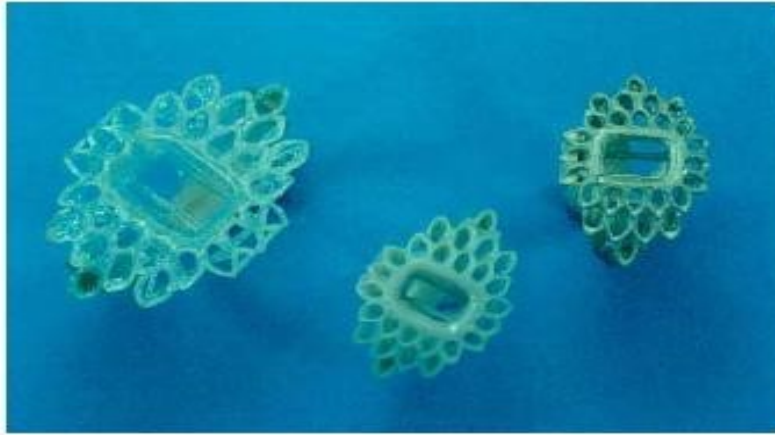


Figure 15. An investment cast silver alloy prototype of a brooch (right), wax pattern created from the silicon rubber molding (center), and the two-time scaled SL built model (left) (Chua et al., 2003).

AM technologies can provide a powerful tool to the jewelers and artists for their work, which will allow them to fabricate unique shaped parts in just a few hours rather than days or weeks. Some systems manufacturers, such as Envisiontec GmbH in Europe and the Solidscape division of Stratasys in the United States, have shown much interest in this particular area and have reported the most work in this area (Chua et al., 2003; Worldwide Guide to Rapid Prototyping [online]).

4.3. Medical Industry

AM technologies have shown their potential in the field of medicine with variety of biomedical applications, as AM coupled with other technology (CT or MRI) have the ability to fabricate complex anatomical parts directly from scanned data (CT images). These AM built models provide a better visualization of a specific anatomy, assist in precise presurgical planning, help the surgeons and medical students to rehearse different surgical procedures realistically, and also function as a communication tool between surgeons and patients (Dhakshyani et al., 2011; Liu et al., 2005; Melchels et al., 2010). SL is being used in biomedical applications (Melchels et al., 2010) to manufacture anatomical implants, tailor-made biomedical devices such as hearing aids and has proven to facilitate and speed up the intensive planning of surgical procedures (Petzold et al., 1999) especially in implant placements and complex surgeries. The SLS process enables the manufacturing of dental implants (Traini et al., 2008), bone scaffolds (Leong et al., 2003) and medical devices (Bertol et al., 2010). Whereas, the FDM process offers the possibility of fabrication of bone models complex shape parts, surgical guides and templates (Liu et al., 2005). In addition the 3DP technology is a suitable method for fabricating precise mandibular reconstruction using bone plates and bone grafts (Hoque, 2011). A study (Zenha et al., 2011) considered 47 mandibular reconstruction cases (between 2003 and 2009) to evaluate the feasibility of applying AM built 3D models in the medical field, concluded that 95.7% of the patients were found to have at least a satisfactory result and the majority (38 out of 47) of patients were in a good and the very good end result categories. Recently this technology was used for the separation of Siamese twins (Tukuru et al., 2008) by performing a precise pre-surgical planning on AM

built models as shown in Figure 16. It is a very significant discovery in the medical sector, which opened the way for creating other complex human organs or tissues with the help of AM technology. But still this technology cannot be used in daily practices due to the presence of current issues like cost, time, suitable material etc. For biomedical applications, further research is required to reduce the overall cost (virtual planning and fabrication cost), development of suitable biomaterials. Table 2 presents some categorized biomaterials.

Table 2. Some categorized biomaterials (Negi et al., 2014)

Metals	Ceramics	Polymers
316L stainless steel	Alumina	Ultra high molecular weight polyethylene
Co-Cr Alloys	Zirconia	Polyurethane
Titanium alloys	Hydroxyapatite	
	Carbon, Calcium phosphate	

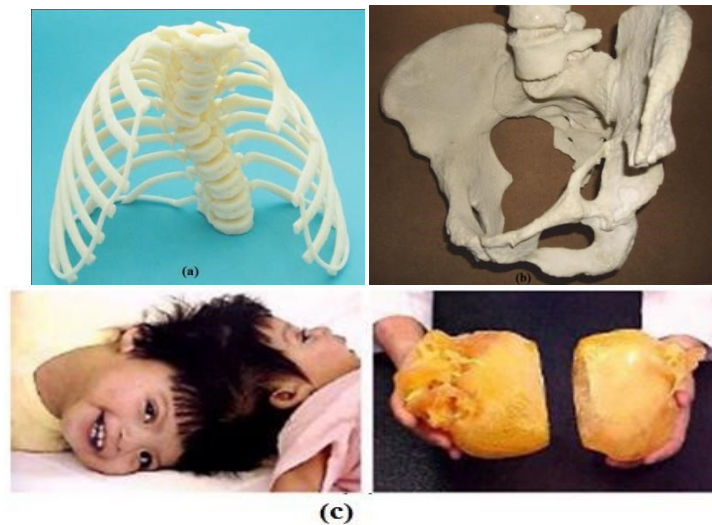


Figure 16. (a) AM model of congenital scoliosis (Hoque, 2011), (b) Pre-surgical planning on the AM model before total hip replacement (Hoque, 2011), (c) Pre-surgical planning on AM built models to separate Siamese twins (Tukuru et al., 2008).

4.4. Architectural Industry

Another interesting field where of this emerging technology has been used is architectural industry. As we know architects usually make their models by performing manual tasks (hand techniques). It consumes a lot of time and sometime it's a hard task to create a complex shape model or to present their ideas in a physical shape. Similar to the jewelry industry, AM technology is capable to create complex shapes quickly and accurately, also it saves time in designing and developing complex shaped architectural models. Exceptional shapes become a reality with the use of AM processes. Stereolithography process is very suitable for the architectural modeling because of its printing resolution and materials availability. Furthermore, designs can be improved easily by simply amending the CAD model. AM technology provides a better resolution than other processes used in architecture (Wong and

Hernandez, 2012; Worldwide Guide to Rapid Prototyping [online]). Models such as presented in Figure 17 can only be fabricated by AM processes.

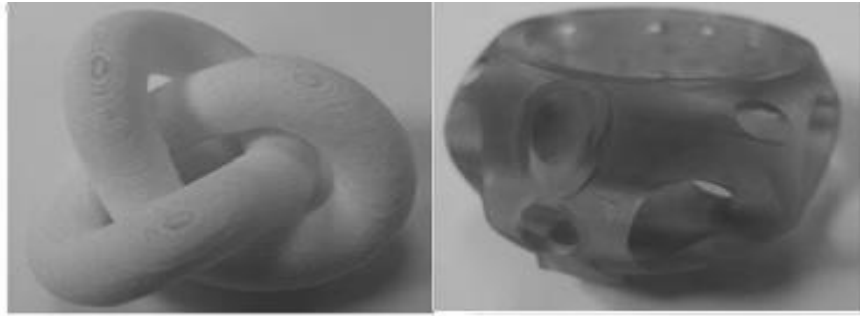


Figure 17. Artistic models fabricated by AM technology (Petrovic et al., 2011).

CONCLUSION

Additive manufacturing technologies are quickly moving from rapid prototyping to rapid manufacturing. Significant improvement (speed, accuracy and material properties) has been noticed in the past decade; hence these technologies are very much capable to fabricate enduser parts with high density and excellent mechanical properties. These technologies provide advantages over the existing manufacturing technologies such as maximum material savings, product customization and require no tooling for fabrication purposes. Due to these reasons these technologies are attracting many industrial sectors such as aeronautic, automotive, artistic and, medical industries.

However, still additive technologies cannot be employed in the manufacturing sector as a standard process due to some issues such as materials to suit applications, long build time, and high cost of the procedure. Therefore, these issues restrict the utilization of AM in certain sectors, where considerable cost savings and higher levels of precision are generally required. Further research and work is required to reduce the overall cost (virtual planning and fabrication cost) of AM process as well as system cost, development of materials and development of AM systems designed specifically for particular applications.

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